

EDOARDO DIANA

Cybersecurity Engineer

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EU citizen • Open to international relocation

PROFESSIONAL SUMMARY

Engineer with a bachelor's degree in Computer and Electronic Engineering and an international double master's degree in Artificial Intelligence and Cybersecurity. Strong Python background and hands-on experience in applied cryptography, hardware security, privacy-preserving computation, and AI-based side-channel analysis, including experimentally evaluated attacks against AES.

TECHNICAL SKILLS

Applied Cryptography: DES, AES, block-cipher fundamentals, Secure Multi-Party Computation, Oblivious Transfer
Hardware & Side-Channel Security: Differential Power Analysis, Correlation Power Analysis, leakage models, profiling
Programming & Analysis: Python, R
AI for Security: TensorFlow/Keras, Multi-Layer Perceptrons, Convolutional Neural Networks, rank-based key evaluation
Systems & Tools: Linux and Docker fundamentals, Git, GitHub

SELECTED CYBERSECURITY PROJECTS

Differential Power Analysis against AES

Individual academic project - Python, NumPy, AES, side-channel analysis

- Extended a provided AES power-analysis script to implement and compare bit-level attacks using Hamming Weight and single-bit leakage models, Pearson correlation, Difference of Means, and t-statistics across non-linear and linear intermediate values.
- Recovered the full 16-byte AES key with 105 traces using correlation; Difference of Means recovered 15 of 16 bytes with approximately 660–700 traces, while attacking the linear AddRoundKey output recovered only 4 of 16 bytes.

Deep Learning for Side-Channel Profiling

Individual academic project - Python, TensorFlow/Keras, ASCAD v1, MLP, CNN

- Implemented profiling side-channel attacks on ASCAD v1 using Multi-Layer Perceptron and Convolutional Neural Network models, processing 700-sample traces and evaluating aligned and desynchronised inputs.
- Compared feature windows, epochs, and batch sizes through rank-based key-recovery experiments; achieved rank 0 for the target byte, while the best MLP configuration recovered up to 4 of 16 bytes in the evaluated project setup.

Secure Multi-Party Computation with Garbled Circuits

Individual academic project - Python, Yao's Garbled Circuits, Oblivious Transfer

- Implemented a two-party Python protocol using Yao's Garbled Circuits and Oblivious Transfer to compute the maximum across Alice's and Bob's integer sets, automatically generating JSON circuits and verifying results against a local computation.

PROFESSIONAL EXPERIENCE

Automation & DevOps Intern - InfoFactory S.r.l.

Udine, Italy

Master's internship

Mar 2026 – Apr 2026

- Developed and published a TypeScript n8n community node for Pumble (GitHub / npm), implementing reusable REST API operations, credential handling, API validation, and technical documentation.
- Prototyped business automation workflows and Docker-based self-hosted environments on a Hetzner VPS, gaining practical experience with Linux, Nginx, DNS, deployment, and server configuration.

Data & Software Engineering Intern - Planeat / Alimentiamoci S.r.l.

Pavia / Milan, Italy

Bachelor's internship and thesis project

Dec 2022 – May 2023

- Built an end-to-end Python pipeline to scrape, normalise, and structure recipe data, then compared six fuzzy-matching approaches and manually validated approximately 1,000 candidate pairs against the company's database.

EDUCATION

International Double Master's Degree in Artificial Intelligence & Cybersecurity

Jul 2026

University of Udine, Italy • University of Klagenfurt, Austria

Bachelor's Degree in Computer and Electronic Engineering

2023

University of Pavia, Italy

LANGUAGES

Italian: Native • **English:** Professional working proficiency (Cambridge B2 First)